

Anritsu claims its BTS Master to be a 21-in-1 instrument that combines cable and antenna analyser, spectrum analyser, power meter and various other functions. Another instrument called Spectrum Master combines a spectrum analyser and an interference analyser together. MT9090A Network Master combines an optical time-domain reflectometer and Ethernet tester.

Size reduction, improved material and more upgrades.

Size optimisation is a focus area in device manufacturing throughout industries and field T&M design is no different. With usage shifting more towards connected technology, desktop equipment are almost out of the scene.

Chavda opined, "The volume of hardware is shrinking in devices. For instance, if you have a cloud-based system, you only need a local display—be it a smartphone or a tablet for field operations. Thus, display devices are reducing. As a result, instrument sizes are shrinking."

Use of superior-quality material is enhancing the resilience of instruments and making these compatible for almost any harsh open field environment. Parag explained, "Major changes have come from the ergonomic side of the instrument. Ruggedness has improved due to the use of high-quality material. This delivers higher capacity for withstanding shock, aberration and ultraviolet exposure. Testing quality has also improved. New features help users to perform measurements quite easily while saving time. Some of these features include improved parking mechanism, holding mechanism and display size."

Market tips: What makes the right device?

When designing a T&M instrument



Clampmeter

for field use, manufacturers should focus on making the device user-friendly. The lower the learning curve for the user, the more the acceptance of the device.

Tripathi elucidated, "Often, field testing instruments are used by less skilled workers. Therefore these must be easy to use and come with online guidance for operation or troubleshooting. Moreover, field testing instruments must have a simple user interface and single button facility. So engineers or managers can

program the instrument in such a way that the field team just needs to connect the instrument and push a single button to complete the measurement."

Along with that, device specifications (frequency, dynamic range, resolution bandwidth, video bandwidth and resolution) should meet majority of the business demands.

Light weight, long battery life and multiple display options are added advantages. The market is flooded with low-quality devices being sold at extremely low prices. These products have a short life and provide inaccurate readings, which can result in injuries for users as well as business losses.

Parag shared his concerns: "Extensive testing equipment require certain levels of safety. Certain instruments from China are being sold in the market at prices as low as ₹ 1200. These are not reliable and also do not meet safety criteria. Products from European manufacturers or Indian companies with foreign collaborations, on the other hand, usually comply with safety standards." Therefore it is imperative for buyers to examine the safety specifications and authenticity of devices under consideration. **EFY**

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